

Luca Brocco

**Junior Data scientist,
Physicist, AI Engineer**

**Graduated in:
LM-17 Physics of Data
UniPd**

Born in Treviso, 31/10/2002

Via Mons. Pasquini 54 - Quarto d'Altino (VE) - 30020

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Skills

Languages: C1 First CAE English certification, Italian native speaker

Coding: Python, R, SQL, C++

Frameworks: PyTorch, Tensorflow, Dask, Docker, AWS Bedrock, Copilot

Knowledge: Data Analysis and Visualization, Machine Learning
Developer, Data Scientist, MLOps, Modeling of Complex Systems

Experience

Data Scientist / AI Engineer Stage @ OMNYS (March - July 2026)

During my Stage at OMNYS I am involved in LLM training and fine tuning using state-of-the-art techniques (SFT, RAG, DPO) for specific agentic tasks in a limited resources scenario.

Data Scientist Stage @ UniPr

(Aug. 2025 - Oct. 2025, Parma)

During my stage at UniParma I joined a research group involved in epidemiology and forecasting of outbreaks caused by vector borne diseases. You may have a look at my [presentation](#).

Academic projects

(Dec. 2024 - Sep. 2026)

I have implemented many advanced Data Analysis solutions for academic projects, making use of most cutting edge technologies. Meteo forecast, Renewable Energy Production, Protein Secondary Structure prediction, parallelization for paper analysis, MAGIC datasets and more were subjects of my past projects.

Formation

Master's Degree in Physics Of Data - 110 w honors/110

Graduated 11/09/2026

(Dic. 2024 - Sep. 2026, University of Padova - Padova, Italy)

My MSc path merges the skills of a Physicist with the ones of a Data Scientist. I learnt how to solve complex problems using advanced techniques and exploiting big Datasets.

Bachelor Degree in Physics (LT-30 Fisica) - 92/110

Graduated 11/12/2024

(Ott. 2021 - Dic. 2024, University of Padova - Padova, Italy)

During my bachelor degree I was able to acquire a deep knowledge of classical and modern Physics, experimental Physics, advanced Mathematics and Statistics and coding in Python and C++

Maturità Scientifica, Liceo Scientifico "L. da Vinci" (Jun. 2021, Treviso)

Thesis, Publications and Portfolio

Master's Thesis: Post-Training Small Language Models for Editorial Compliance Checking: How Output Structure Inverts the Ranking of Alignment Methods

Summary: The Thesis explores training of an in-production agentic system for automated compliance checking of editorial articles in a constrained resources scenario. The central finding is the influence of output structure on the performance of models and preference alignment algorithms on the task.

The content of the Thesis is currently submitted at EACL 2027.

Bachelor's Thesis: Caratterizzazione tramite spettroscopia ottica di un plasma a pressione atmosferica

Summary: Active carbon filters are commonly used in water PFAS cleaning, but their disposal is currently performed via thermic processes that scatter such particles in the atmosphere. The Thesis explores the possibility of using a double-frequency cold plasma for cleaning such filters instead.

Project portfolio: <https://lucabrocco.github.io/>